

Visualisation 2 submission

Github repository link: <https://github.com/33159483/FIT3179/tree/main/Vis2>

Visualisation link: <https://33159483.github.io/FIT3179/Vis2/>

5 Design sheets:

[https://github.com/33159483/FIT3179/blob/main/Vis2/5DesignSheets/5 Design Sheet Vis2.pdf](https://github.com/33159483/FIT3179/blob/main/Vis2/5DesignSheets/5%20Design%20Sheet%20Vis2.pdf)

Discussion Questions

The domain, the why and the who of the visualisation

- Domain → The visualisation aims to depict disasters which have occurred within Australia, with a particular focus upon instances between 1900 and 2025.
- Why → The reasoning behind the visualisation is for possible preventative actions to which may result from further projections of the data at hand. The disaster factors, trends, seasonalities and geographical locations can lead to mitigation efforts to limit the instances of these catastrophic events, to which have high fatalities attached.
- Who → This visualisation would be relevant for policy makers and those involved within infrastructure. Given some of the events have more correlations with certain geographical locations it would be useful for local councils to have differing standards of housing given high probabilities of events such as cyclones or flooding.

What: The data (sources, authors, relevance, creation process, etc.)

- The data was collected from two government agencies. In particular the visualisation projected data which came from the Australian Government's National Emergency Agency and the Attorney-Generals Department, both of which provided relevant datasources, which kept disasters monitored and up to date. The second dataset held valuable geographical location data which was used within the symbol map. The first dataset was used throughout the rest of the dataset.

- The data types:
 - Quantitative: Fatality counts, latitude, longitude
 - Categorical
 - Ordinal: Start dates, end dates
 - Nominal: Disaster types, states, regions, disaster events

Data sources can be found below

<https://knowledge.aidr.org.au/collections/australian-disasters/>

<https://data.aurin.org.au/dataset/au-govt-agd-disaster-events-impact-location-na>

How: Give a rationale for choosing the specific idioms and explain how they help the users to achieve their tasks. Describe any special features of your visualisation, such as custom-built elements.

- Successfully Added Features:
 - To combat the unsuccessful features below I have incorporated interactive filters throughout the visualisation to depict state trends across temporal and categorical variables.
 - I thought the additional photos will can be found behind the subheadings provides relevant data, through depictions of disasters which are made apparent through the visualisation.
 - I believe the idioms used throughout the visualisation successful indicate disasters which have occurred throughout Australia. The heat-map allows for seasonal factors to be made apparent, along with the bar chart and stacked bar chart to depict trends in specific disaster types and finally the symbol map and table provides useful information of notable events and their geographical location.
- Unsuccessful features:
 - Unsuccessful attempts at adding tree diagrams, radar charts and Sankey chart. I believe these charts would have provided further complexities to the visualisations. Especially the sankey chart will grants an ease depiction to see how the types of disasters flow through each state.